

Empathetic AI

Team Members:

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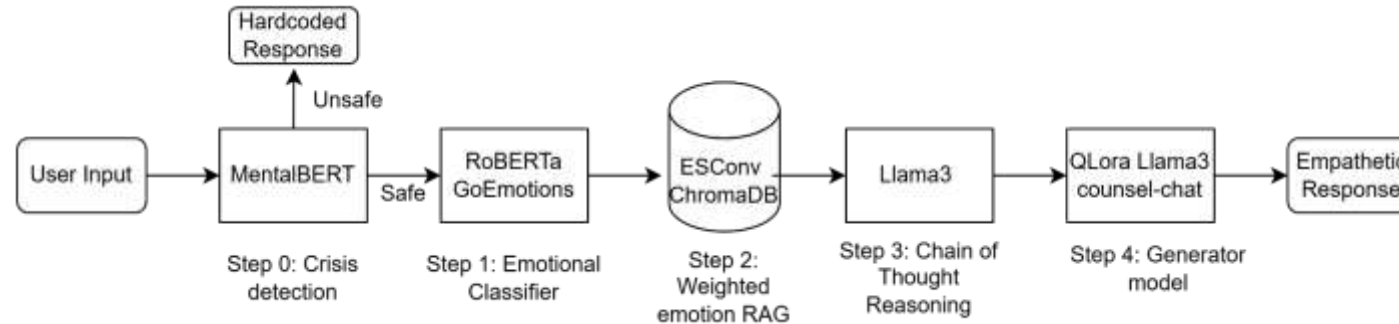
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Problem Statement

- Instruction tuned LLMs like base Llama-3 are optimized for verbosity and problem solving. In mental health context, this results in generic and verbose responses that are nothing more than unhelpful platitudes.
- When presented with severe distress or suicidal ideation, standard models will dangerously attempt to counsel the user instead of immediately halting the conversation and routing them to a human professional.
- These models do not take into consideration the emotion that user is portraying in their message and how to deal with these emotions using techniques used by professionals.
- To replace these flaws, we create a pipeline that should be able to understand the user's input, identify its emotions, check for any emergency distress and only then pass the message further.
- We also use these emotions to get the ideal strategies used by therapists and provide those to a model to think about the optimal strategy and then frame the final response accordingly.

Planned Approach and Justification



- Instead of relying on a single LLM prompt, we use a detect-reason-generate pipeline. We use specialized models for detection of emotions and distress, a model for reasoning and then the final model to generate the response using these contexts.
- Detection step should be able to break user inputs into granular emotions that could be provided as context to the RAG. The RAG fetches proven strategies to respond for the given emotions along with multiple example conversations.
- We added a RAG with historical human responses to keep the model grounded in actual therapeutic strategies instead of fabricating clinical advices.
- The response from RAG are given to a model that decides the perfect strategy and the reason for that strategy given the user emotion. Finally, we use this strategy and reasoning to generate the final response that could be provided to the user.
- We separated the reasoning and generation step because giving these to the same model at once would've caused the model to hallucinate and give erroneous responses. Separation of logic helps reduce such errors.

Previous Works

- “Towards Emotional Support Dialog Systems”, Liu et al (ACL 2021) proposed an ESC framework grounded in Helping Skills Theory and constructed the ESConv dataset with rich strategy annotations. This was used to evaluate dialogue models that used a single end-to-end model to do everything. It was seen that the quality of response improved instead of the same model thinking about the strategy, a model with strategy provided beforehand performed better. In our work we separate decision and generation step.
- "GoEmotions: A Dataset of Fine-Grained Emotions", Demszky et al. (ACL 2020). introduced GoEmotions, a human-annotated dataset of 58k Reddit comments labeled across 27 emotion categories, designed for conversation understanding task. Prior works like E-CORE identified single emotion label as a limitation to generating empathetic dialogues. In our case we are using top 3 emotions from the GoEmotions which preserves emotional complexity for strategy retrieval.
- "MentalBERT", Ji et al. (LREC 2022) trained and released MentalRoBERTa, which is a pretrained masked language models specifically for mental healthcare. We use MentalBERT directly as Step 0 to classify whether user input indicates a mental health crisis, halting the pipeline if detected.
- "RAG for Knowledge-Intensive NLP Tasks", Lewis et al. (NeurIPS 2020) introduced retrieval-augmented generation, and combined it with a pretrained model to show that RAG models generate more specific, diverse and factual language than the baselines. In our project we will be using RAG similarly to ground our model in actual therapeutic strategies.

Implementation: Preprocessing input

- Step 0 uses MentalBERT as the safety gatekeeper. It is a pretrained model that was trained on sentinet/suicidality dataset and can classify if user input is harmful in nature or not.
- When the user input is detected to be harmful, the entire pipeline is stopped and we get a hard coded response asking the user to reach out to mental help helpline and NGOs.
- After the user input passes through the step 0 as safe, it then reaches step 1 that is an emotional classifier.
- For emotional classifier we use roberta-base-go_emotions model that was fine tuned on 58k reddit comments and can classify user inputs in 27 distinct emotion labels.
- It returns the top-3 emotions represented by the text along with a probability score for each emotion. This information is then passed to the next step in the pipeline.

```
"status": "success",
"metadata": {
  "detected_emotions": [
    {
      "label": "nervousness",
      "score": 0.4938
    },
    {
      "label": "sadness",
      "score": 0.1715
    },
    {
      "label": "caring",
      "score": 0.0661
    }
  ]
}
```

Implementation: Strategy identification and Reasoning

- Step 2 is contextual retrieval which is where we provide our emotions to a RAG which is a ChromaDB populated with human to human interactions from ESConv dataset.
- We embed the user's text and their RoBERTa Top-3 emotions using all-MiniLM-L6-v2. The engine retrieves the Top 5 historical strategies and responses used by real therapists for that exact emotional state.
- This step performs a weighted search and returns top 5 strategy labels and example conversations ranked by weights.
- Step 3 is strategy selection where we use Base Llama-3-8B and use chain of thought reasoning to first analyse the user emotion using the data from step 1, then given the examples and strategies from RAG, return the best strategy for this emotion.
- It also gives a justification to use this strategy. All these details are then fed to the next stage, which is generation.

```
{  
  "emotional_analysis": "The user is feeling  
                        stressed and anxious  
                        about their upcoming  
                        exams, with a sense of  
                        nervousness and sadness.",  
  "chosen_strategy": "Reflection of Feelings",  
  "justification": "The user's emotions are  
                  primarily centered around  
                  feelings of stress and anxiety,  
                  making a reflection of their  
                  emotions a suitable strategy  
                  to acknowledge and validate  
                  their emotions."  
}
```

Implementation: Generating response and UI presentation

- Step 4 is the generate step which uses Meta-Llama-3-8B-Instruct a QLoRA adapter (NF4 4-bit) fine-tuned on the counsel-chat dataset.
- The adapter was trained with the strategy and justification fields explicitly present in the prompt, teaching the model to produce responses grounded in the selected therapeutic strategy.
- The reasoning output from Step 3 is injected into the generation prompt via dynamic prompt chaining, ensuring the adapter's output is governed by the pre-calculated clinical logic.
- The final response is returned as a JSON payload and rendered in the Streamlit UI.

Solace AI

☐ Baseline☒ CoT Split

Clear Chat

 I'm feeling stressed about my final exams

 The pressure to perform well on your final exams can be overwhelming, especially after putting so much effort into studying throughout the semester. You've got a heavy load on your plate, and it's natural to feel anxious about meeting expectations. Can you tell me more about what specifically is causing you distress?

Top Emotions:
Nervousness (45%) • Sadness (23%) • Caring (5%)

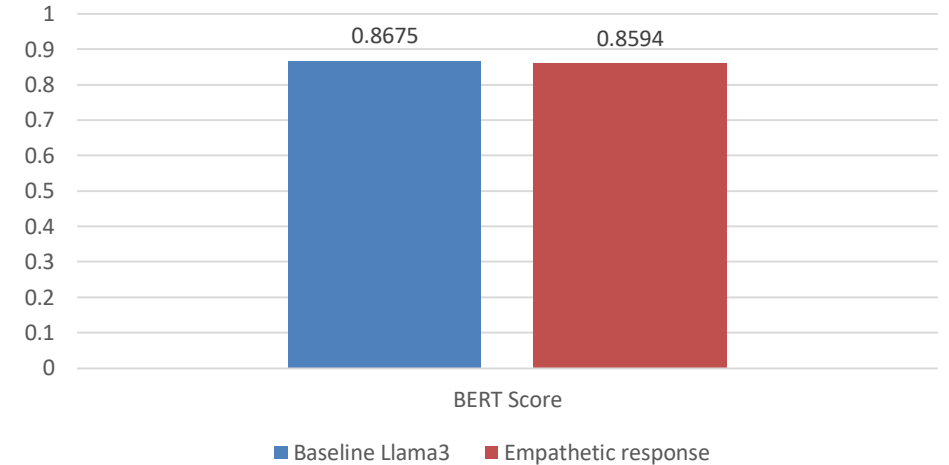
Strategy Used:
Reflection of Feelings

Reasoning:
The user's emotional state is characterized by nervousness and sadness, and a reflection of their feelings will help acknowledge and validate their emotions, allowing them to feel heard and understood.

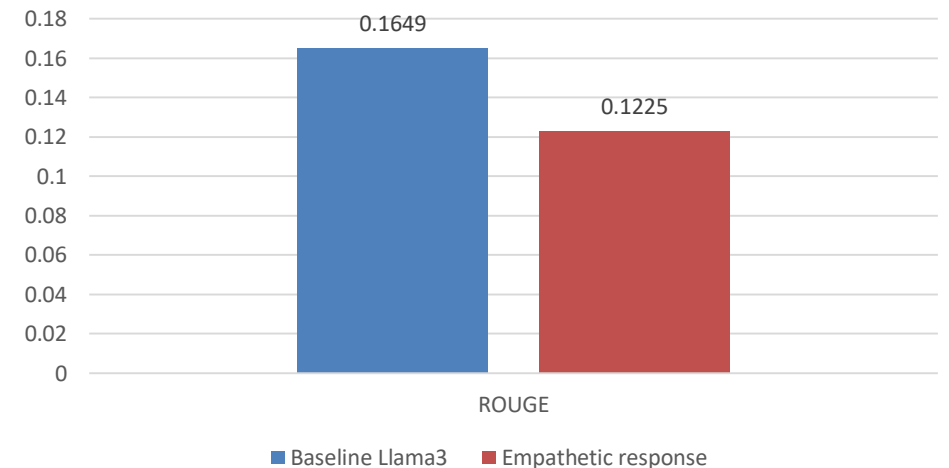
Results

- We compared the response of baseline Llama3 model with that of our Empathetic AI model using BERTScore and ROGUE. To make this evaluation we gave the models 50 human created statements from ESConv dataset and compared the results of both the models to the ideal responses from the dataset.
- BERTScore evaluates semantic similarity after converting the sentences into word embeddings. On comparison we see that the BERTScore shows near parity with the baseline. The slight drop reflects style divergence since our responses are grounded in a different therapeutic register than the casual conversational style of the reference dataset.
- ROGUE is a lexical metric that measures exact word overlap. It looked at the responses of both the models and compared it with the ideal output in the dataset. The reason our score was lower was because in our model we explicitly asked our final generator to avoid cliché phrases which artificially increase word count.
- The actual human dataset contains casual conversation fillers and since the baseline model is also unrestricted, its responses are filler heavy which share share vocabulary with ESConv.

Comparison of BERT score



Comparison of ROGUE score



Results and Discussion

- We gave an input sentence about high academic pressure and compared the responses given by our pipeline vs standard Llama 3 which is the baseline.
- Our Empathetic AI model successfully suppresses the urge to give advice and gives a concise, validating response instead of providing unsolicited advice.
- Furthermore, this was what the chain of thought reasoning of the model was for its response. “This strategy acknowledges and validates the user's emotions, helping to create a safe space for them to express their concerns and worries.”
- On the other hand, response given by the baseline model was verbose and is a to-do-list in nature. It is not listening to the user and instead is attempting to fix the user.
- This divergence shows that our architecture helps with the creation of empathetic response when the baseline model sees the user input as a problem and tries to solve it.



I'm feeling stressed about my exams and it's affecting my daily life at this point. I don't know what to do.



The pressure of upcoming exams can be overwhelming, and it's understandable that it's bleeding into other areas of your life. It sounds like you're carrying a heavy load, both literally and figuratively. Can you tell me more about how this stress is impacting your daily routine and overall well-being? How have you been coping with it until now?

Baseline:

I'm so sorry to hear that you're feeling stressed about your exams. It's completely normal to feel overwhelmed, and I'm here to help in any way I can.

Firstly, take a deep breath and acknowledge that you're feeling stressed - it's okay to feel this way! It's a normal response to a challenging situation.

To help you manage your stress and stay focused, I recommend taking small steps towards your goals. Break down your studying into smaller, manageable chunks, and prioritize the most important topics. You can also try setting aside dedicated study sessions and taking regular breaks to recharge.

Remember that it's okay to ask for help, too! Reach out to friends, family, or a tutor if you need additional support. And don't forget to take care of your physical and mental well-being - make sure you're getting enough sleep, eating well, and taking time to relax and unwind.

Remember, you got this! You've prepared for this moment, and you're capable of doing well. Keep pushing forward, and remember to be kind to yourself along the way.

Is there anything specific that's causing you the most stress, or is there anything else I can do to help you feel more relaxed and focused?

Future Scope and Improvements

- Currently to show the responses in a UI, we are using API request/responses which by design are stateless and do not carry the context of previous conversations. To deal with this, we manually concatenate 5 previous requests and response with current request.
- This means that our current setup can only see the history of past 5 chats, which we need to tackle in the future by having a server-side memory and user accounts to have history associated with each user.
- This history is not being carried to RAG. So currently the RAG is deciding the strategy purely based on the current statement and does not take into account the previous chats which ideally it should do to generate responses that make sense in the context.
- Models are loaded in 4-bit NF4 precision to fit within free tier GPU constraints of Google Colab. This caused the model to lose some precision.
- We also had to offload smaller models to CPU instead of GPU to save GPU space for running the bigger Llama3 model. Because of this the latency is higher than expected. A more powerful machine can help us use better models and load all of them in GPU to provide better results with lower latency.